

Bug Fix Impact Analysis

METIMMOX-1 Economic Evaluation

Ben Geisler

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Table of contents

1	Summary	1
2	Metadata Comparison	2
2.1	Global Parameters	2
3	Base Case Results Comparison	3
3.1	Net Monetary Benefit Comparison	3
4	PSA Results Comparison	4
4.1	Efficiency Frontier Comparison	5
4.2	PSA Visual Analysis	6
4.2.1	PSA Scatter Plots (Side-by-Side)	6
4.2.2	PSA Scatter Plots (Overlaid)	7
5	Comparison Across Recent Issues	8
5.1	Available Snapshot Pairs	8
5.2	Multi-Issue Impact Summary	8
6	Summary	9

1 Summary

This report compares cost-effectiveness analysis results before and after addressing GitHub issue #70. The comparison assesses whether bug fixes or model changes affected results.

Current Comparison:

- **Issue:** #70 - PSA uses inconsistent methods for control vs. biomarker strategy predictions
- **Baseline:** Commit 998dacd (2025-11-21 09:38:06)
- **Fixed:** Commit 605ff90 (2025-11-21 11:46:32)

2 Metadata Comparison

Table 1: Metadata Comparison

Item	Baseline	Fixed
Git Commit	998dacd	605ff90
Timestamp	2025-11-21 09:38:06	2025-11-21 11:46:32
R Version	R version 4.5.1 (2025-06-13 ucrt)	R version 4.5.1 (2025-06-13 ucrt)
Issue Number	70	70

2.1 Global Parameters

Table 2: Global Parameter Comparison

	Parameter	Baseline	Fixed	Changed
WTP	WTP	51,000	51,000	No
time_horizon	time_horizon	520	520	No
n_samples	n_samples	5,000	5,000	No
n_sim	n_sim	5,000	5,000	No
cl	cl	0.01923077	0.01923077	No
dr	dr	0.04	0.04	No
DSA_mult	DSA_mult	0.2	0.2	No
USE_BOTH_MODELS	USE_BOTH_MODELS	0	0	No

3 Base Case Results Comparison

Table 3: Base Case Incremental Cost-Effectiveness Comparison

Strategy	Incremental Cost		Incremental Effect		Baseline
	Baseline Inc. Cost (€)	Fixed Inc. Cost (€)	Baseline Inc. Effect (QALYs)	Fixed Inc. Effect (QALYs)	
control	NA	NA	NA	NA	
crp	34,244.12	NA	0.1112	NA	
tlr	NA	NA	NA	NA	
tmb_braf	NA	41,540.41	NA	0.0252	

3.1 Net Monetary Benefit Comparison

Table 4: Net Monetary Benefit Comparison at WTP = €51,000

	Strategy	Baseline (€)	Fixed (€)	Difference (€)
1	control	70,257.69	73,356.74	3,099.06
2	crp	41,685.57	36,372.76	-5,312.81
4	tmb_braf	24,963.72	33,103.52	8,139.80
3	tlr	-10,359.72	2,163.83	12,523.55

Optimal Strategy: Unchanged (control)

4 PSA Results Comparison

Table 5: PSA Cost Results (n = 5,000 simulations)

Strategy	Baseline Mean (2.5%; 97.5%)	Fixed Mean (2.5%; 97.5%)
control	€20,936.34 (€15,783.79; € 27,129.21)	€20,747.36 (€15,676.15; € 26,817.10)
crp	€54,118.74 (€40,595.71; € 69,484.76)	€54,856.22 (€16,137.87; €152,119.33)
tlr	€85,905.64 (€61,363.13; €115,121.66)	€86,892.87 (€15,140.97; €163,776.53)
tmb_braf	€64,950.69 (€47,458.10; € 85,506.50)	€65,757.95 (€17,309.20; €155,120.39)

Table 6: PSA Effects Results (n = 5,000 simulations)

Strategy	Baseline Mean (2.5%; 97.5%)	Fixed Mean (2.5%; 97.5%)
control	1.6999 (1.1101; 2.3569)	1.8435 (1.3388; 2.0390)
crp	1.6977 (1.1276; 2.2533)	1.8837 (0.9958; 3.4041)
tlr	1.6230 (1.0532; 2.1950)	1.7947 (0.5740; 3.2621)
tmb_braf	1.7942 (1.1067; 2.5413)	1.9835 (1.0243; 3.3431)

Table 7: PSA ICER Results (n = 5,000 simulations)

Strategy	Baseline Mean ICER	Fixed Mean ICER
control	—	—
crp	—	€849,081/QALY
tlr	—	—
tmb_braf	€466,674/QALY	€321,509/QALY

Note: ICERs calculated from PSA mean costs and effects as point estimates.

4.1 Efficiency Frontier Comparison

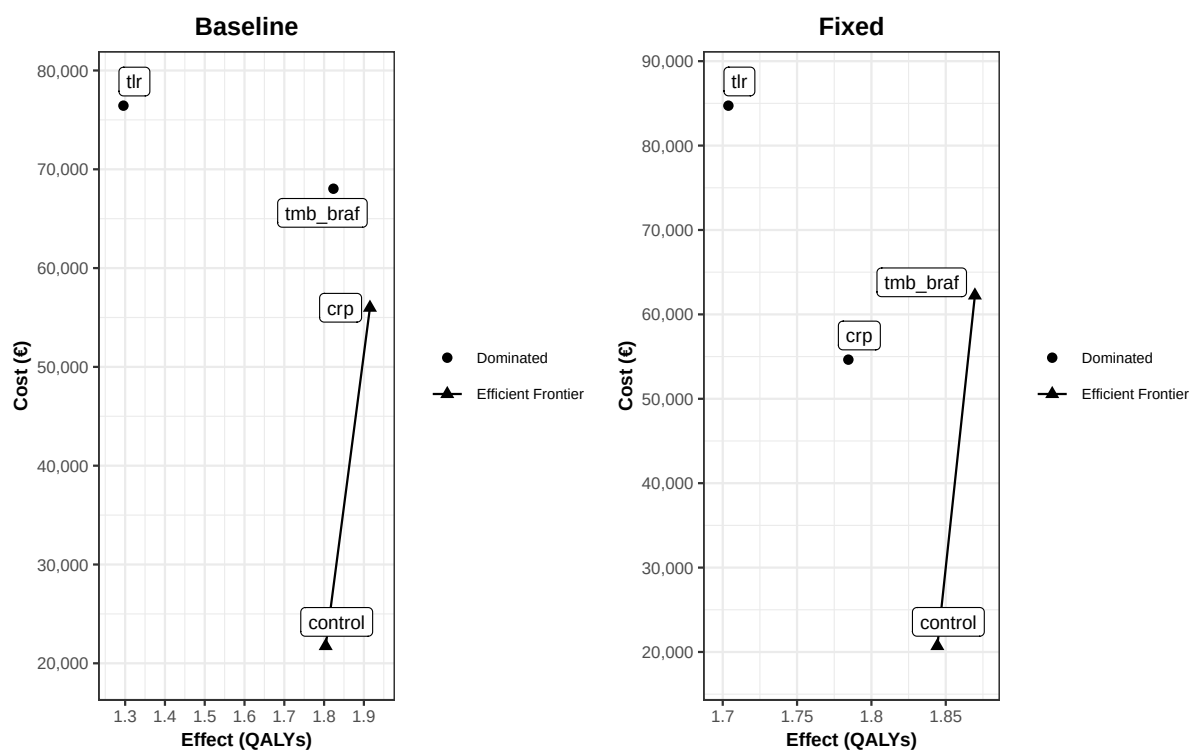


Figure 1: Cost-effectiveness efficiency frontier comparison using dampack. Baseline (left) and fixed (right) strategies are shown with the efficiency frontier connecting non-dominated strategies. Dominated strategies appear off the frontier line.

4.2 PSA Visual Analysis

4.2.1 PSA Scatter Plots (Side-by-Side)

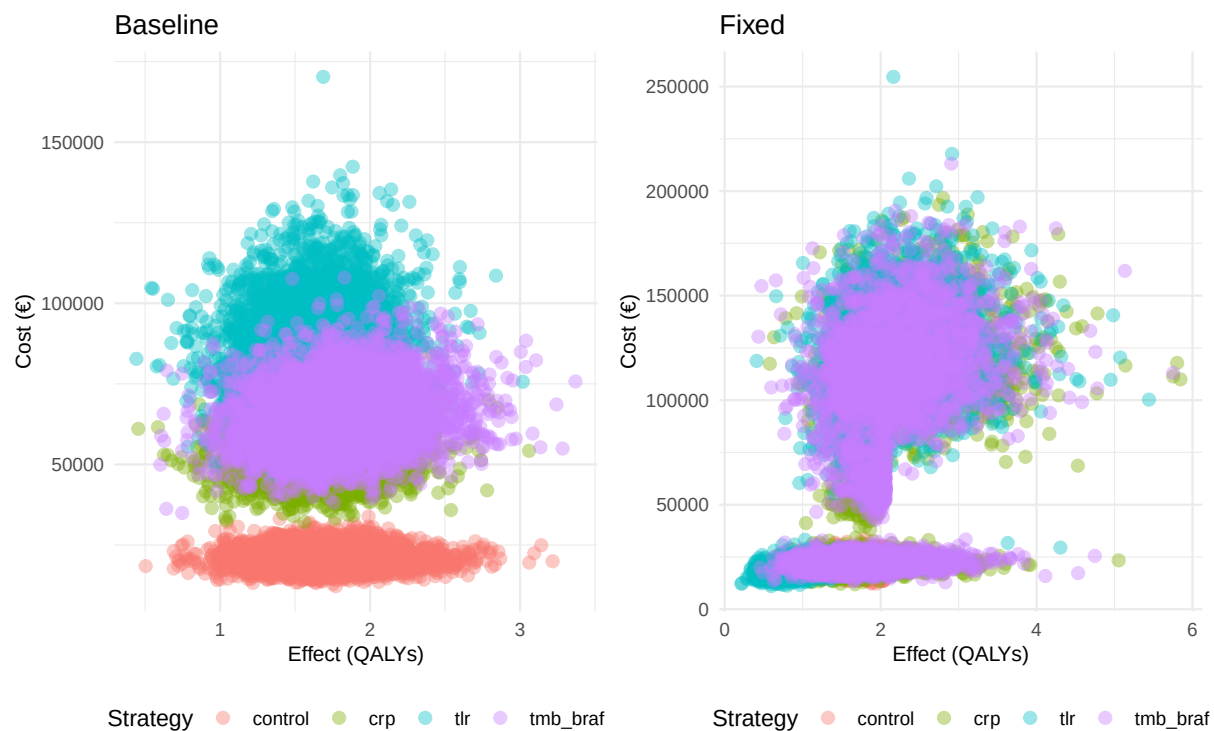


Figure 2: Cost-effectiveness plane showing PSA simulations baseline (left) and fixed (right)

4.2.2 PSA Scatter Plots (Overlaid)

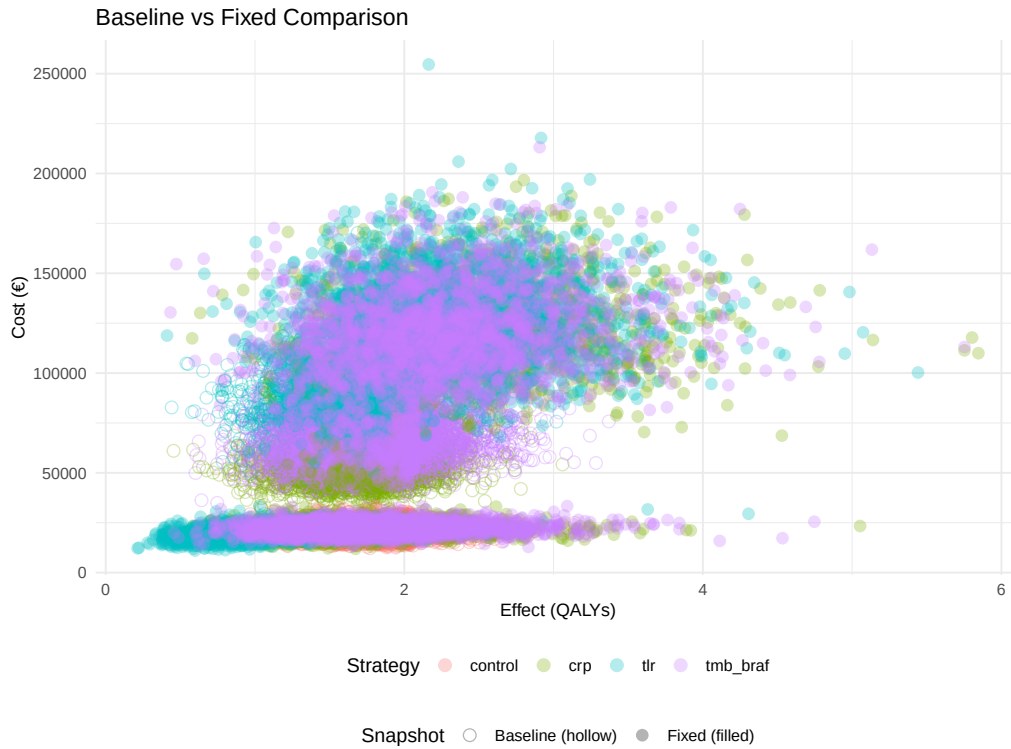


Figure 3: Overlaid cost-effectiveness plane comparing baseline and fixed PSA simulations. This plot overlays all PSA simulation results from both snapshots on a single plane, where each point represents one PSA iteration for a given strategy. Hollow circles indicate baseline simulations, while filled circles indicate fixed simulations. Colors distinguish strategies, while shapes distinguish snapshots. The overlay allows direct visual comparison of how the probabilistic distributions changed after the bug fix: if the fix was purely methodological with no impact on central estimates, baseline and fixed points would overlap completely, whereas systematic shifts in the point clouds indicate changes in underlying model behavior.

5 Comparison Across Recent Issues

5.1 Available Snapshot Pairs

Table 8: All Available Snapshot Pairs

	Issue	Title	Baseline Commit	Fixed Commit	Status
6	#70	PSA uses inconsistent methods for control vs. biomarker strategy predictions	998dacd	605ff90	Current
5	#68	[P4] Include Weibull model for reference	4665f93	998dacd	
1	#42	[P0] Critical Bug: Incorrect death transition calculation in calculate_outcomes	607b539	cadf81a	
4	#67	[P4] Move snapshot folder	f3674e8	d5d1356	
2	#62	[P4] Move cached file	d046c7c	f3674e8	
3	#64	[P3] impact of bug fixes	8681d1e	8681d1e	

5.2 Multi-Issue Impact Summary

This table summarizes the impact of bug fixes across the most recent issues:

Table 9: Summary of Bug Fix Impacts (5 Most Recent Issues)

Issue	Title	Opt. (BL)	Opt. (Fix)	Chg?	Max Cost Δ (€)	Max Effect Δ (QALYs)	Max ICER Δ (€/QALY)
#70	PSA uses inconsistent methods for control vs. biomarker strategy predictions	control	control	No	8,290.42	0.4081	-Inf
#68	[P4] Include Weibull model for reference	control	control	No	0.00	0.0000	0
#42	[P0] Critical Bug: Incorrect death transition calculation in calculate_outcomes.R	control	control	No	0.00	0.0000	0
#67	[P4] Move snapshot folder	control	control	No	0.00	0.0000	0
#62	[P4] Move cached file	control	control	No	0.00	0.0000	0

6 Summary

The comparison for issue #70 shows:

- **Optimal Strategy:** Unchanged (control)
- **Maximum Cost Change:** €8,290.42 (substantial)
- **Maximum Effect Change:** 0.408 QALYs (substantial)
- **Maximum ICER Change:** €-Inf/QALY (minor)
- **Parameter Changes:** None detected

Interpretation: Changes exceed typical thresholds and should be carefully reviewed to ensure they align with the intended bug fix.

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